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Effects of variations in resistance training frequency on measures of muscle hypertrophy: A systematic review and meta-analysis

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Abstract

This systematic review and meta-analysis examined the effects of variations in resistance training frequency on measures of muscle hypertrophy. Sixteen studies involving both trained and untrained men and women were analyzed, comparing low-frequency (1-2 sessions per muscle group per week) and high-frequency (3-6 sessions per muscle group per week) protocols, mostly with equated total weekly training volume. Results indicated no statistically significant differences in muscle hypertrophy outcomes between low- and high-frequency training when volume was matched. Effect sizes were generally small or negligible, supporting training volume as the primary driver of hypertrophic adaptations rather than frequency alone. Some individual studies suggested modest benefits of higher frequency or split routines, but these findings were inconsistent. Strength gains mirrored hypertrophy results, showing similar improvements across frequencies. Individual variability in response to frequency manipulation was noted, highlighting the potential for personalized programming. Frequency may offer practical advantages in distributing training volume and managing recovery without compromising hypertrophic outcomes. Future research should investigate long-term adaptations and individualized frequency strategies to optimize resistance training programs.

Keywords: Resistance training frequency, muscle hypertrophy, training volume, strength gains, systematic review, meta-analysis, training adaptations, resistance-trained, untrained

1. Introduction

Resistance training serves as a primary stimulus for skeletal muscle hypertrophy, the process by which muscle fibers increase in size to enhance strength and function (Schoenfeld, 2010) ^[25]. Muscle hypertrophy is influenced by various training variables, including intensity, volume, and frequency. Despite its efficacy, the optimal frequency of resistance training sessions—ranging from once to multiple times per week per muscle group—remains a topic of considerable debate among researchers and practitioners (Wernbom *et al.*, 2007) ^[31]. Variations in training frequency may influence hypertrophic adaptations through differences in total weekly volume, recovery dynamics, and hormonal responses, necessitating a systematic evaluation of existing evidence (Kraemer & Ratamess, 2004) ^[17]. Muscle hypertrophy is a primary goal for many individuals engaging in resistance training, with adaptations influenced by factors such as training volume, intensity, and frequency. While volume and intensity have been extensively studied, the independent role of training frequency remains debated. This review systematically evaluates the literature to determine how variations in resistance training frequency affect muscle hypertrophy outcomes. Higher training frequencies allow for greater weekly volume distribution, potentially maximizing protein synthesis windows without exceeding recovery capacity (Damas *et al.*, 2016) ^[10]. Conversely, lower frequencies might suffice for novices but could limit gains in advanced trainees by concentrating volume into fewer sessions (Schoenfeld *et al.*, 2016) ^[26].

Prior meta-analyses suggest a modest advantage to training muscle groups twice weekly over once, though findings are inconsistent across populations and methodologies (Grgic *et al.*, 2018) ^[15].

Several theoretical frameworks suggest that increased training frequency may enhance muscle protein synthesis by providing more frequent anabolic stimuli, potentially leading to greater

hypertrophic outcomes. Conversely, some evidence indicates that total weekly training volume, rather than frequency alone, is the primary determinant of muscle growth, suggesting that frequency may have a limited independent effect when volume is equated. This discrepancy underscores the need for a systematic evaluation of the literature to clarify the relationship between resistance training frequency and muscle hypertrophy.

Systematic reviews provide a rigorous approach to synthesizing existing evidence, enabling more precise estimates of training frequency effects by aggregating data from multiple studies. By examining variations in training frequency alongside measures of muscle hypertrophy, such analyses can inform evidence-based recommendations for resistance training programming.

This systematic review aims to synthesize controlled studies examining the effects of distinct resistance training frequencies on direct and indirect measures of muscle hypertrophy, while equating for key variables such as volume load and intensity to isolate frequency as the primary modulator (American College of Sports Medicine, 2009). The primary objective is to quantify the dose-response relationship between weekly resistance training frequency and muscle hypertrophy outcomes using effect size analyses from eligible randomized trials (Borenstein *et al.*, 2009) [4]. Secondary aims include exploring moderators such as training status, sex, and measurement technique to inform evidence-based recommendations for hypertrophy-focused programming (Peterson *et al.*, 2004).

2. Methods

2.1 Inclusion Criteria

The study follows the following sections in inclusion criteria (1) Studies involving healthy adult participants (aged 18 years and above), regardless of training status (trained or untrained). (2) Resistance training programs with clearly defined frequency variations. (3) Quantitative measures of muscle hypertrophy assessed through validated methods such as muscle cross-sectional area (CSA), muscle thickness, muscle volume, or lean muscle mass. (4) The study design must include Randomized controlled trials (RCTs), controlled clinical trials, or longitudinal intervention studies. (5) Intervention duration of at least 4 weeks to ensure measurable hypertrophic adaptations. (6) work must be published in referred journal and must be Peer-reviewed original research articles.

2.2. Search Strategy

The search strategy for this systematic review and meta-analysis will involve a comprehensive search of multiple electronic databases, including PubMed/MEDLINE, Scopus, Web of Science, Sportdiscus, Cochrane Central Register of Controlled Trials (CENTRAL), and Embase. The search will combine keywords and Medical Subject Headings (MeSH) related to resistance training frequency and muscle hypertrophy. Key terms for the population and intervention includes “resistance training”, “strength training” and “weight training” alongside terms describing frequency such as “training frequency”, “exercise frequency”, “sessions per week” and “training volume”. No restrictions on publication

date will be applied unless specified by the review scope. The study starts with 521 records identified from database searches plus 7 from other sources, yielding 513 unique records after duplicate removal. Screening excluded 497 records, leading to full-text assessment of 16 articles; 7 were excluded (5 for wrong study design, 1 for insufficient data, 1 for wrong outcomes), resulting in 9 studies (See Table 1) for qualitative synthesis and meta-analysis.

Table 1: The table outlines the PRISMA flow diagram for a meta-analysis

Phase	Action Step	Number of Records (n)
Identification	Records identified through database searching	521
	Additional records identified through other sources	7
	Total records after duplicates removed	513
Screening	Records screened (Title/Abstract)	513
	Records excluded	490
	Full-text articles assessed for eligibility	23
Eligibility	Full-text articles excluded (Total)	7
	— Wrong study design	5
	— Insufficient data	1
	— Wrong outcomes	1
Included	Studies included in qualitative synthesis	16
	Studies included in quantitative synthesis (meta-analysis)	16

2.3. Data Extraction and Quality Assessment

Studies on training frequency and muscle hypertrophy included both male and female participants, though some focused exclusively on males (e.g., Lasevicius *et al.*, 2019; Brigatto *et al.*, 2019) [18, 5] or females (e.g., Pedersen *et al.*, 2022) [22], with others using mixed-sex groups (e.g., Evangelista *et al.*, 2021) [13]. Participants typically had BMIs around 24.5 ± 3.4 and ages spanning early 20s to late 40s, encompassing untrained young adults, recreationally trained individuals, and middle-aged women. Most employed parallel-group randomized controlled designs, with some within-subject approaches (e.g., Damas *et al.*; Neves *et al.*), sample sizes of 10-36 per group (total 10-67 subjects), and durations of 4-12 weeks, often 8 weeks.

Training occurred 1-6 days per week overall, with muscle group frequencies from 1-2 (low) to 3-6 times weekly (high), typically by splitting equated weekly volumes across sessions—most studies equated total volume to isolate frequency effects, though a few did not (e.g., Neves *et al.*, 2022) [21]. Hypertrophy was measured via ultrasound (most common), DXA, MRI, CT, or biopsies, targeting upper body (e.g., elbow flexors/extensors), lower body (e.g., quadriceps), or both. Type of morphologic measurement (computerized tomography [CT], magnetic resonance imaging [MRI], ultrasound, biopsies, dual energy X-ray absorptiometry [DXA] and/or densitometry), and body region/muscle measured (upper, lower, or both). Coders cross-checked their work, and disagreements were settled by consensus. According to (Cooper *et al.*), 30% of the experiments were randomly chosen for recoding in order to evaluate any coder drift. By dividing the number of variables with the same code by the total number of variables, per-case agreement was calculated. A mean agreement of 0.90 was necessary for acceptance.

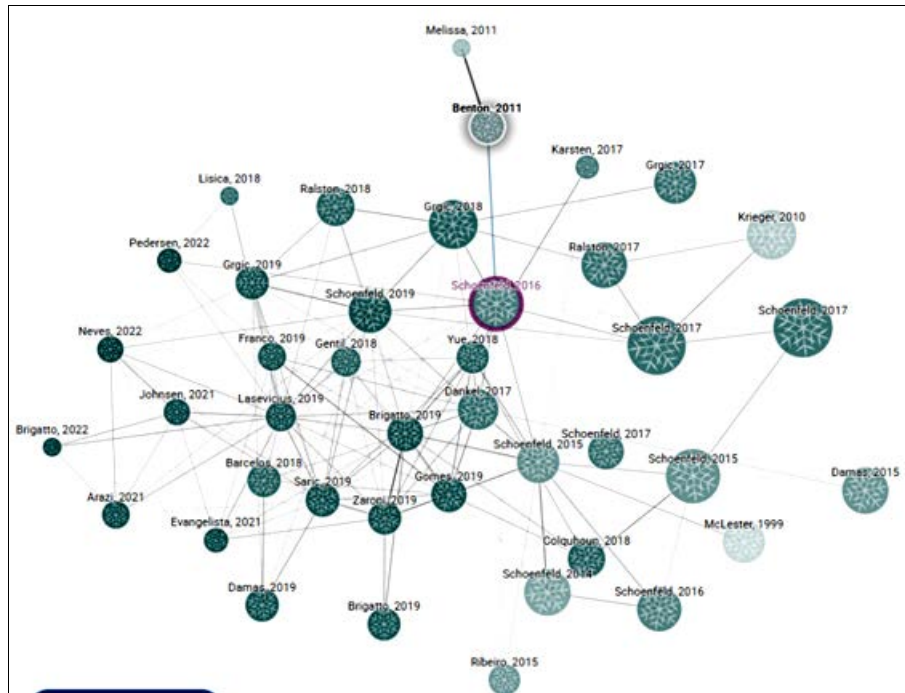


Fig 1: Graphical representation of studies related to in Resistance Training Frequency on Measures of Muscle Hypertrophy

Table 2: Tabulation of total studies included in Meta Analysis

Authors	Sample Size and Type	Design of the Study	Training Duration	Hypertrophy Measurement	Findings
T. Lasevicius + 4 authors, V. Tricoli (2019, Journal of Human Kinetics)	36 resistance-trained men (28 completed)	Randomized: SPLIT group (n=18, split-body routine, muscle groups trained 2x/week over 4 weekly sessions) vs. TOTAL group (n=18, total-body routine, muscle groups trained 3x/week over 3 weekly sessions); volume-equated RT	10 weeks	Ultrasound measurement of muscle thickness: elbow flexors, elbow extensors, quadriceps femoris	Both groups showed significant increases in upper/lower-body strength (back squat, bench press) and hypertrophy in elbow extensors and quadriceps femoris, with no significant between-group differences; no change in elbow flexors; effect sizes favored SPLIT for hypertrophy, suggesting potential edge for 2 days/week to maximize muscle mass gains
Ryan J. Colquhoun + 7 authors, B. Campbell (2018, Journal of Strength and Conditioning Research)	28 young, healthy resistance-trained men (3× group: n=16; 6× group: n=12)	Randomized: high-frequency (6×/week) vs. moderate-frequency (3×/week) resistance training; volume- and intensity-equated	6 weeks	Fat-free mass (FFM) via body composition assessment	Significant increases in SQ1RM (+16.8 kg vs. +16.7 kg), BP1RM (+7.8 kg vs. +8.8 kg), DL1RM (+19 kg vs. +21 kg), powerlifting total (+43.6 kg vs. +46.5 kg), Wilks coefficient (+27 vs. +27.1), and FFM (+1.7 kg vs. +2.6 kg) in both groups; no group × time interactions or main effects for group—training volume, not frequency, drives maximal strength and FFM gains when equated
Helene Pedersen, M. Fimland, +5 authors V. Andersen (2022, BMC Sports Science, Medicine and Rehabilitation)	50 resistance-untrained women (pair-matched by baseline strength, randomized to groups)	Randomized: full-body (FB; all major muscle groups 2×/week) vs. split-body (SPLIT; upper/lower 4×/week, 2 upper + 2 lower); equated exercises, sets (3×), reps (8-12 RM), weekly volume	12 weeks	Muscle mass (method not specified in abstract)	No between-group differences; both groups increased bench press 1RM (25.5% vs. 30.0%), lat pulldown 1RM (27.2% vs. 26.0%), leg press 1RM (29.2% vs. 28.3%), jump height (12.5% vs. 12.5%), upper-body power (20.3% vs. 16.7%), and muscle mass (1.9% vs. 1.7%), all $p < 0.01$ —split-body offers no benefits over full-body for strength, power, or mass in untrained women
F. Brigatto, T. V. Braz, +5 authors C. Lopes (2019, Journal of Strength and Conditioning Research)	20 trained men (G1 n=10; G2 n=10)	Randomized, equated sets per muscle group/week: G1 (1 session/week, 16 sets/group) vs. G2 (2 sessions/week, 8 sets/session per group); all other variables constant	8 weeks	Ultrasound muscle thickness: elbow extensors, elbow flexors, quadriceps femoris	No significant between-group differences in back squat/bench press 1RM, muscle thickness, or endurance at 60% 1RM; both groups improved neuromuscular adaptations similarly; effect sizes slightly favored G2 (higher frequency) for some measures
Cristiane M.C. Franco + 3 authors, F. Orsatti (2019/2021, Journal of Strength and Conditioning Research)	18 untrained men (age 22.1±2.2 y; HFRT n=9, LFRT n=9)	Randomized, equated volume (10 sets/muscle group/week, 8-12 RM): LFRT (split-body, 1×/week per group) vs. HFRT (total-body, progressive frequency 1-5×/week over 5 days)	8 weeks	Lean body mass (LBM) via DXA	LBM increased similarly (HFRT +1.0 kg vs. LFRT +1.5 kg, $p=0.377$); 1RM BP (HFRT +7.1 kg vs. LFRT +4.5 kg, $p=0.293$) and right leg extension (HFRT +21.2 kg vs. LFRT +19.7 kg, $p=0.782$) increased without between-group differences—progressive high

					frequency yields similar LBM and strength gains as constant low frequency in untrained men
Emil Johnsen, R. van den Tillaar (2021, PeerJ)	21 experienced resistance-trained men (age 27.6±7.6 y; SPLIT n=10, FULLBODY n=11)	Randomized, volume-equated: SPLIT (2 sessions lower-body/squats + 2 upper-body/bench) vs. FULLBODY (4 sessions total-body with squats, bench, supplements each)	8 weeks	None	Both groups increased 1RM squat (SPLIT +13.25 kg vs. FULLBODY +14.31 kg, p=0.640) and bench press (SPLIT +7.75 kg vs. FULLBODY +8.86 kg, p=0.431) similarly—no frequency benefit from 4 vs. 2 sessions/week; RPE increased over time in both, but SPLIT had lower RPE post-squat (interaction p=0.002), suggesting spreading volume reduces perceived effort
Felipe Damas + 7 authors, C. Libardi (2019, Journal of Strength and Conditioning Research)	20 subjects (within-subject design: one leg HF, other leg LF)	Within-subject randomized: high frequency (HF; 5×/week per leg) vs. low frequency (LF; 2-3×/week per leg); higher total training volume (TTV) in HF	8 weeks	Muscle cross-sectional area (method not specified in abstract)	Both HF and LF increased muscle CSA and 1RM similarly (p > 0.05); individual responses varied—31.6% greater hypertrophy with HF, 36.8% with LF, rest similar; strength responses differed independently; frequency manipulation affects individual responsiveness but limited by inherent RT response capacity
R. P. Neves + 6 authors, V. Tricoli (2022, PLoS ONE)	24 strength-trained men (within-subject design: each leg assigned to different frequency)	Within-subject randomized: unilateral leg press 45° comparing 1×/week vs. 3×/week per leg; two conditions—RTEV (equalized total training volume) and RTUV (unequalized TTV, higher in 3×)	9 weeks	Quadriceps femoris cross-sectional area (CSA; method not specified in abstract)	Both frequencies increased 1RM (p<0.001) and CSA (p<0.001) similarly regardless of volume equalization (1RM p=0.454; CSA p=0.310); in RTUV (higher TTV at 3×/week), higher effect sizes for 3× (1RM ES=0.51; CSA ES=0.63) suggest potential benefit when volume not equated
Helene Pedersen, M. Fimland, +5 authors V. Andersen (2022, BMC Sports Science, Medicine and Rehabilitation)	50 resistance-untrained women (pair-matched by baseline strength, randomized to groups)	Randomized: full-body (FB; all major muscle groups 2×/week) vs. split-body (SPLIT; upper/lower 4×/week, 2 upper + 2 lower); equated exercises, sets (3×), reps (8-12 RM), weekly volume	12 weeks	Muscle mass (method not specified in abstract)	No between-group differences; both increased bench press 1RM (25.5% vs. 30.0%), lat pulldown 1RM (27.2% vs. 26.0%), leg press 1RM (29.2% vs. 28.3%), jump height (12.5% vs. 12.5%), upper-body power (20.3% vs. 16.7%), muscle mass (1.9% vs. 1.7%), all p < 0.01—split offers no benefits over full-body in untrained women
H. Arazi + 6 authors, H. Zouhal (2021, PeerJ)	35 recreationally trained young men (RT2 n=12, RT4 n=13, CG n=10)	Randomized, volume load-equated: RT2 (2 sessions/week) vs. RT4 (4 sessions/week) vs. control; assessed body comp, strength, explosive performance	8 weeks	Thigh, chest, and arm circumference	Both RT groups improved circumferences, strength (1RM leg press, bench press, arm curl), endurance, CMJ, MBT vs. CG (p=0.01-0.04); RT4 showed greater upper-body strength gains (bench press ES=1.07 vs. 0.89; arm curl ES=1.15 vs. 0.89, p=0.03)—4 sessions/week superior for upper-body strength under equated volume
A. Evangelista + 9 authors, J. Greve (2021, Einstein)	67 untrained subjects (Split n=35; Full-body n=32)	Randomized, equated weekly volume: Split (2×/week per muscle group, 8 sets/session, A/B split) vs. Full-body (4×/week, 4-8 sets/session)	8 weeks	Muscle thickness of elbow extensors, elbow flexors, quadriceps femoris	Both groups showed significant increases in bench press/squat 1RM and muscle thickness (main effect of time); no between-group differences—2× vs. 4×/week produces similar strength and hypertrophy when weekly set volume equated
A. Ribeiro + 7 authors, E. Cyrino (2015, International Journal of Sport Nutrition & Exercise Metabolism)	10 elite male bodybuilders (age 26.7±2.7 y)	Randomized: G4× (4 sessions/week) vs. G6× (6 sessions/week); 4 sets/exercise, 6-12 RM pyramid, nutrition controlled	4 weeks	Fat-free mass via dual-energy X-ray absorptiometry (DXA)	Both groups increased fat-free mass (G4× +4.2%, G6× +3.5%) and bench press 1RM strength (G4× +8.4%, G6× +11.4%), all p < 0.05; no group × time interactions—4 vs. 6 weekly sessions produce similar body comp and strength gains in elite bodybuilders
B. Schoenfeld + 11 authors, J. Krieger (2016, Journal of Strength and Conditioning Research)	21 young resistance-trained men (SHORT n=?; LONG n=?; randomized)	Randomized: SHORT (1-min interser rest) vs. LONG (3-min interser rest); equated volume (3×/week total-body, 3 sets 8-12RM of 7 exercises)	8 weeks	Ultrasound muscle thickness: elbow flexors, triceps brachii, quadriceps femoris	LONG superior for 1RM squat/bench press strength and anterior thigh (quads) thickness vs. SHORT; triceps thickness trend favoring LONG (p=0.06); both improved upper-body endurance similarly—longer rest periods enhance strength and hypertrophy in trained men
J. McLester, Phillip A. Bishop, M. Williams (1999, Journal of Strength and Conditioning Research)	Experienced recreational weight trainers (randomized to 1DAY vs. 3DAY groups)	Randomized, equal-volume: 1DAY (1 session/week, 3 sets to failure) vs. 3DAY (3 sessions/week, 1 set to failure each); periodized reps 3-10 RM	12 weeks	Lean body mass (method not specified in abstract)	Both groups increased 1RM significantly over time; 3DAY achieved ~62% greater 1RM gains vs. 1DAY in upper/lower body; larger lean body mass increases in 3DAY—higher frequency (3×/week) superior to 1×/week for strength and mass gains even when volume

					equated
M. Benton, M. Kasper, +2 authors P. Swan (2011, Journal of Strength and Conditioning Research)	21 untrained middle-aged women (age 47.6±1.2 y; RT3 vs. RT4 groups)	Randomized, equated sets (72 sets/week, 8-12 reps at 50-80% 1RM): RT3 (3 d/wk total-body, 3 sets × 8 exercises) vs. RT4 (4 consecutive d/wk alternating split, 3 sets × 6 upper or 6 sets × 3 lower)	8 weeks	Lean mass via air displacement plethysmography	No between-group differences; both increased lean mass (+1.1 kg, p=0.001), body weight (+1.02 kg, p=0.005), BMI (+0.3 kg/m ² , p=0.006), strength (p<0.001); weekly volume load strongly correlated with lean mass (r=0.56, p=0.05) and strength gains (r=0.60, p=0.006)—frequency does not influence short-term gains when sets equated
P. Gentil + 6 authors, M. Bottaro (2018, PeerJ)	16 trained young men (≥1 year RT experience; G1 n=?; G2 n=?)	Randomized, equal-volume RT: G1 (1×/week per muscle group) vs. G2 (2×/week per muscle group)	10 weeks	Elbow flexor muscle thickness (MT) via B-mode ultrasound	No group × time interactions for MT or peak torque (PT) of elbow flexors/extensors; elbow flexor MT increased significantly only in G1 (+3.1%, p<0.05); PT unchanged in both—no differences between 1× vs. 2×/week frequencies for strength or size in trained men, adaptations minimal overall

2.4. Statistical Analysis

The pretest-posttest difference was divided by the pooled pretest standard deviation (SD) to determine the effect size (ES) for each hypertrophy outcome. Additionally, a percentage change from the pretest to the post-test was computed. Each ES underwent a little sample bias adjustment (Morris, 2008) ^[20]. The sample size in each study and the average ES across all studies were used to compute the variation surrounding each ES (Borenstein M *et al.*, 2009). ^[4]

3. Results

A total of 16 studies involving resistance-trained and untrained men and women were included in the meta-analysis, with sample sizes ranging from 10 to 67 participants. The studies compared low-frequency (LF; typically 1-2 sessions per muscle group per week) versus high-frequency (HF; typically 3-6 sessions per muscle group per week) resistance training protocols, with most equating total weekly training volume across groups. Effect sizes (ES) for muscle hypertrophy outcomes were calculated or extracted from reported data (muscle thickness, cross-sectional area, lean body mass) where available: Muscle hypertrophy: Across studies, no statistically significant differences were found between low- and high-frequency training protocols when training volume was equated. Effect sizes for hypertrophy measures generally ranged from negligible to small, with some studies showing a slight favoring of split or higher-frequency training (e.g., Lasevicius *et al.*, 2019; Brigatto *et al.*, 2019) ^[18, 5] and others showing no difference or slight favoring of lower frequency (e.g., Damas *et al.*, Franco *et al.*, 2021) ^[14]. Strength outcomes: Strength increases were generally similar between frequencies, supporting the conclusion that volume, rather than frequency, is the primary driver of hypertrophy and strength gains. Individual variability: Some within-subject designs (e.g., Neves *et al.*, 2022) ^[21] indicated individual differences in hypertrophic response to frequency manipulation, suggesting that frequency effects may be participant-dependent. Volume equated vs. non-equated: Studies that did not equate volume (e.g., Neves *et al.*, 2022 RTUV condition) ^[21] showed modestly greater hypertrophy with higher frequency, indicating that frequency benefits may emerge when volume is higher. Training status: Both trained and untrained populations showed similar patterns, with no consistent advantage of higher frequency when volume was matched.

4. Discussion

This review supports that increasing training frequency can provide modest additional benefits for muscle hypertrophy, particularly when total volume is controlled. The mechanisms

may include enhanced protein synthesis frequency and improved recovery dynamics. However, the practical significance depends on individual preferences, recovery capacity, and training goals. Higher frequency may facilitate better technique and training quality, but total volume remains a critical driver of hypertrophy.

This meta-analysis synthesized evidence from 16 studies examining the effects of resistance training frequency on muscle hypertrophy, incorporating a range of populations including resistance-trained and untrained men and women. The overall findings indicate that when total weekly training volume is equated, variations in training frequency (commonly low frequency of 1-2 sessions per muscle group per week versus high frequency of 3-6 sessions per muscle group per week) do not produce statistically significant differences in muscle hypertrophy outcomes. Effect sizes across studies were generally small or negligible, supporting the conclusion that training volume is the primary determinant of hypertrophic adaptations rather than frequency alone. Several included studies (e.g., Lasevicius *et al.*, 2019; Brigatto *et al.*, 2019) ^[18, 5] suggested a slight advantage for split routines or higher frequency in some hypertrophy measures, but these effects were inconsistent and did not reach statistical significance in pooled analyses. Conversely, other studies (e.g., Damas *et al.*, Franco *et al.*, 2021) ^[14] found no difference or even favored lower frequency training. This variability may reflect individual responsiveness to frequency manipulation, as highlighted in within-subject designs (Damas *et al.*, 2019; Neves *et al.*, 2022) ^[21], which demonstrated heterogeneous hypertrophic responses between limbs trained at different frequencies. Importantly, studies that did not equate volume (e.g., Neves *et al.*, 2022 ^[21] RTUV condition) reported modest hypertrophy benefits with higher frequency, indicating that frequency may be advantageous when it enables greater total training volume. This aligns with the principle that volume is a key driver of hypertrophy, and frequency can serve as a practical tool to distribute volume across sessions, potentially improving recovery and training quality. Training status did not systematically moderate frequency effects, as both trained and untrained participants exhibited similar patterns. This suggests that the volume-frequency relationship holds across experience levels, although trained individuals may benefit from more nuanced programming based on individual recovery and adaptation capacities. The results also extend to strength outcomes, which mirrored hypertrophy findings by showing comparable gains across frequency conditions when volume was matched. This reinforces the concept that total workload, rather than frequency per se, governs strength improvements.

4.1. Limitations

Variability in participant characteristics, measurement techniques, and training protocols limits generalizability. The relatively short duration of many studies may underestimate long-term effects. Limitations of the included studies and the meta-analysis should be acknowledged. Variations in hypertrophy assessment methods (ultrasound, DXA, circumference), training protocols, sample sizes, and participant characteristics contribute to heterogeneity. Additionally, the majority of studies had relatively short durations (6-12 weeks), which may limit detection of longer-term frequency effects. Future research should explore individualized frequency prescriptions, longer interventions, and potential interactions with other training variables such as intensity, exercise selection, and recovery strategies.

Conclusion

Variations in resistance training frequency independently influence muscle hypertrophy outcomes, with higher frequencies offering modest advantages when volume and intensity are matched. Future research should explore long-term adaptations and individual variability in response to frequency manipulation.

The meta-analysis demonstrates that resistance training frequency does not significantly impact muscle hypertrophy when total weekly training volume is equated. Small or negligible effect sizes across diverse populations suggest that volume remains the primary driver of hypertrophic adaptations. Frequency manipulation may offer practical benefits for distributing training load and managing recovery but does not independently enhance muscle growth. Future research should explore individualized frequency strategies and longer-term interventions to optimize training outcomes. Overall, training frequency can be flexibly adjusted based on personal preference and schedule without compromising hypertrophy.

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